

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

ORDER NO. 5-00-265

WASTE DISCHARGE REQUIREMENTS
FOR
CITY AND COUNTY OF SAN FRANCISCO
MOCCASIN SEWAGE TREATMENT PLANT
TUOLUMNE COUNTY

The California Regional Water Quality Control Board, Central Valley Region (hereafter Board), finds that:

1. The City and County of San Francisco (hereafter Discharger) submitted a Report of Waste Discharge (RWD), dated 8 September 2000, for the treatment and disposal of domestic sewage and maintenance area wastewater generated at the Moccasin Powerhouse and the Hetch Hetchy Water and Power Administrative complex.
2. The facility is on Assessor's Parcel Number (APN) 57-180-05, and is owned and operated by the Discharger. The effluent storage pond and reclamation area are on APN 735-342-44, and are also owned and operated by the Discharger.
3. The wastewater treatment plant is located near the north end of the Moccasin Reservoir dam adjacent to Moccasin Creek, approximately 1,000 feet north of Highway 49, and 2,000 feet southwest of Highway 120. The facility is in Sections 27 and 34, T1S, R15E, MDM&M, as shown on Attachment A, which is attached hereto and made part of the Order by reference. The effluent storage pond and sprayfield are located on the south side of Highway 49, approximately one-half mile west of the treatment plant, in Section 28, T1S, R15E, MDM&M, as shown on Attachment A.
4. Order No. 75-55, adopted by the Board on 21 March 1975, prescribes requirements for a large septic tank discharging to a leachfield. This Order is neither adequate nor consistent with the current plans and policies of the Board.
5. Wastewater originates from sources within a complex that contains 44 residences, a maintenance shop, the powerhouse (domestic waste only), and the administrative offices. Wastewater from the maintenance shop is primarily domestic. The wastewater treatment plant serves approximately 50 to 60 permanent residents, 70 to 80 workers per day using the facility during the week, and 25 to 30 workers from remote locations that periodically use the facilities.
6. The current dry weather flows are approximately 20,000 gallons per day (gpd), with peak wet weather flows reaching approximately 70,000 gpd. The occurrence of peak wet weather flows that are significantly higher than average dry weather flows suggests that there is a significant amount of inflow and infiltration into the wastewater collection system. Minimal growth is expected for the facility, and so it is not anticipated that the wastewater flows will increase significantly in the foreseeable future.

7. Two storm water drop inlets at the fuel island and the drop inlet for the vehicle wash pad are routed through an oil/water separator before the wastewater flows to the treatment plant. The drop inlets from the fuel island are scheduled to be decommissioned as part of an upgrade and repair of the Moccasin wastewater system, for which the contract should be completed by 15 February 2001. When the work is completed, there will be no stormwater or other flows from the fuel island to the sewage collection and treatment system.
8. Wastewater is conveyed from the complex to the treatment plant through a gravity collection system, with ultimate delivery by means of a lift station. The lift station has an exterior power connection receptacle that can be used with a portable generator during power failures, and an alarm system should failure occur.
9. Based on the Discharger's Report of Waste Discharge, the average influent strength is reported to be:

Biological Oxygen Demand -	75 mg/l
Total Suspended Solids -	100 mg/l
Total Dissolved Solids -	250 mg/l
Total Nitrogen as N -	20 mg/l

10. The treatment system consists of an extended aeration package plant, with disposal by means of spray irrigation. Certified operators perform treatment plant operations and maintenance. The treatment plant consists of three components: an aeration basin with two rotary blowers for air supply, a sedimentation basin for solids removal, and a chlorine contact tank. The contact tank also serves as a wet well for pumping treated effluent to the holding pond and spray field area. The treatment plant has a standby generator with a manual start-up and switchover control in the event of power failure.
11. Based on the Discharger's Report of Waste Discharge, the average effluent strength is reported to be:

Biological Oxygen Demand -	<5 mg/l
Total Suspended Solids -	10 mg/l
Total Coliform -	50 MPN/100 ml
pH -	6.5

12. The Discharger is currently constructing an equalization/overflow structure consisting of two 18,000 gallon tanks, located near the treatment plant. The tanks will be operated in parallel, in series, or individually, as needed. The purpose of the structure is to intercept excessive flows through the treatment plant. The normal function will be to extend chlorine contact time in the contact chamber and provide an equalization chamber during periods of excessive flow.

13. Effluent from the treatment plant is pumped from the chlorine contact tank via a force main to the effluent storage pond. Effluent pumps are operated by float switches.
14. An oil/water separator/carbon filter treatment system is immediately adjacent to the pond. This is used to treat wastewater collected in sumps at three powerhouses (Moccasin, Kirkwood, and Holm). Most of the wastewater collected in the powerhouse sumps is infiltration water and floor drain water with some oil. The wastewater is trucked to the treatment system, where it is pumped into an approximately 15,000-gallon tank before being batch treated with the oil/water separator and two carbon filters in series. After treatment, the water is pumped to the pond.
15. The effluent storage pond is in a seasonal drainage, with an earthen dam to increase capacity. Upslope storm water is diverted under the pond and dam by means of a 48-inch culvert. The total capacity of the pond is approximately 720,000 gallons. The capacity of the pond while maintaining two feet of freeboard is approximately 480,000 gallons. The Discharger has no record of overflows from the pond; however, a water balance for the effluent storage and disposal system indicates that there is not enough storage capacity in the pond to assure that two feet of freeboard is maintained during the wet season. According to the Discharger, the pond is lined with clay. However, no information is available regarding the thickness, integrity, or hydraulic conductivity of the liner.
16. Effluent is removed from the pond using vertical turbine pumps that are activated by time clocks, and sprayed onto the 10-acre sprayfield. During the dry season, the pond's elevation is maintained with a freeboard of 6 to 7 feet. However, during the wet season, the pond is pumped down to accommodate rainfall.
17. A diversion ditch directs runoff from the spray field. Runoff can be directed back to the pond, or can be routed around the pond and into the seasonal drainage below the pond.
18. The majority of solids produced by the facility is in the form of sludge, which is dried prior to disposal on the sprayfield. Approximately six cubic yards of dried sludge (biosolids) and a small amount of screenings have been disposed of per year on the sprayfield. The Discharger has indicated that, beginning in December 2000, sludge from the plant will be disposed at an approved landfill appropriate for sludge disposal.
19. The potable water supply source is snowmelt runoff from the Hetch Hetchy basin. No wells are reported in the vicinity.
20. The flow at this wastewater treatment plant is less than 1.0 mgd and therefore the Discharger is not required to apply for a stormwater NPDES permit
21. The Basin Plan encourages reclamation.
22. The California Department of Health Services has established statewide reclamation criteria in Title 22, California Code of Regulations, Section 60301, et. seq. (hereafter

Title 22), for the use of reclaimed water, and has developed guidelines for specific use. These requirements implement the reclamation criteria in Title 22.

23. Surrounding land uses are primarily rural agricultural.
24. The Board adopted a Water Quality Control Plan, Fourth Edition, for the Sacramento River and San Joaquin River Basins (hereafter Basin Plan), which contains water quality objectives for waters of the Basins. These requirements implement the Basin Plan.
25. Surface water drainage is to Moccasin Creek, a tributary of New Don Pedro Reservoir.
26. The beneficial uses of sources to New Don Pedro Reservoir are municipal and domestic supply; agricultural supply; power generation, contact and noncontact recreation; canoeing and rafting; warm and cold freshwater habitat, and wildlife habitat.
27. The beneficial uses of underlying groundwaters are municipal, industrial, and agricultural supply.
28. In order to determine compliance with the groundwater limitation contained herein, these WDRs contain a time schedule for development and implementation of a groundwater monitoring program.
29. The Board has considered anti-degradation pursuant to State Board Resolution No. 68-16 and finds that not enough data exists to determine whether this discharge is consistent with those provisions. Therefore, this Order provides a timeline for data collection to determine whether the discharge will cause an increase in groundwater constituents above that of background levels. If the discharge is causing such an increase, then the Discharger may be required to cease the discharge, reline the pond, implement source control, change the method of disposal, or take other action to prevent groundwater degradation.
30. The action to update waste discharge requirements for this facility is exempt from the provisions of the California Environmental Quality Act (CEQA), in accordance with Title 14, California Code of Regulations (CCR), Section 15301.
31. This discharge is exempt from the requirements of *Consolidated Regulations for Treatment, Storage, Processing, or Disposal of Solid Waste*, as set forth in Title 27, CCR, Division 2, Subdivision 1, Section 2005, et seq., (hereafter Title 27). The exemption pursuant to Section 20090(b), is based on the following:
 - a. The Board is issuing waste discharge requirements,
 - b. The discharge complies with the Basin Plan, and
 - c. The wastewater does not need to be managed according to Title 22 CCR, Division 4.5, and Chapter 11, as a hazardous waste.

32. Section 13267(b) of California Water Code provides that: "In conducting an investigation specified in subdivision (a), the regional board may require that any person who has discharged, discharges, or is suspected of discharging, or who proposes to discharge within its region, or any citizen or domiciliary, or political agency or entity of this state who has discharged, discharges, or is suspected of discharging, or who proposes to discharge waste outside of its region that could affect the quality of the waters of the state within its region shall furnish, under penalty of perjury, technical or monitoring program reports which the board requires. The burden, including costs of these reports, shall bear a reasonable relationship to the need for the reports and the benefits to be obtained from the reports."
33. The Board has notified the Discharger, and interested agencies and persons of its intent to prescribe waste discharge requirements for this discharge and has provided them with an opportunity for a public hearing and an opportunity to submit their written views and recommendations.
34. The Board, in a public meeting, heard and considered all comments pertaining to the discharge.

IT IS HEREBY ORDERED that Order No. 75-55 is rescinded and the City and County of San Francisco, its agents, successors, and assigns, in order to meet the provisions contained in Division 7 of the California Water Code and regulations adopted thereunder, shall comply with the following:

A. Discharge Prohibitions:

1. Discharge of wastes to surface waters or surface water drainage courses is prohibited.
2. Bypass or overflow of untreated or partially treated waste is prohibited.
3. Neither the treatment nor the discharge shall cause a nuisance or condition of pollution as defined by the California Water Code, Section 13050.
4. The discharge shall not cause the degradation of any water supply.
5. Discharge of waste classified as hazardous, as defined in Sections 2521(a) of Title 23, CCR, Section 2510, et seq., (hereafter Chapter 15, or 'designated', as defined in Section 13173 of the California Water Code, is prohibited.
6. Surfacing of wastewater outside and downgradient of the pond is prohibited.
7. Discharge of untreated or partially treated waste to groundwater is prohibited.
8. Overflow of wastewater from the effluent storage pond is prohibited.

9. The discharge of any wastewater other than that from domestic sources, the drop inlets for storm water described in Finding No. 7, or the oil water separator/carbon filter system described in Finding No. 14 is prohibited.
10. Application of wastewater to areas different than that described in Finding No. 3 is prohibited.
11. The discharge of wastes from recreational vehicles into the wastewater treatment plant is prohibited.

B. Discharge Specifications:

1. The average dry weather discharge shall not exceed 25,000 gallons per day (gpd). Average monthly wet weather flows shall not exceed 70,000 gallons per day.
2. Objectionable odors originating at this facility shall not be perceivable beyond the limits of the wastewater treatment and disposal areas.
3. As a means of discerning compliance with Discharge Specification No.2, the dissolved oxygen content in the upper zone (1 foot) of the storage pond shall not be less than 1.0 mg/l.
4. Effluent in the storage pond shall not have a pH of less than 6.5 or greater than 8.5.
5. The Discharger shall operate all systems and equipment to maximize treatment of wastewater and optimize the quality of the discharge.
6. The pond shall be managed to prevent the breeding of mosquitoes. In particular,
 - a. An erosion control program should assure that small coves and irregularities are not created around the perimeter of the waste surface.
 - b. Weeds shall be minimized through control of water depth, harvesting, and/or herbicides.
 - c. Dead algae, vegetation, and debris shall not accumulate on the water surface.
7. The Discharger's wastewater treatment system shall be designed, constructed, operated, and maintained to prevent inundation or washout due to floods with a 100-year return frequency.
8. Prior to 1 December 2002, the freeboard in the storage pond shall never be less than one foot as measured vertically from the water surface to the lowest point of overflow. After this date, the freeboard shall never be less than two feet.
9. The wastewater ponds shall have sufficient capacity to accommodate allowable wastewater flow, design seasonal precipitation, and ancillary inflow and infiltration.

Design seasonal precipitation shall be based on total annual precipitation using a return period of 100 years, distributed monthly in accordance with the historical rainfall patterns.

10. On or about 1 October each year, available pond storage capacity shall at least equal the volume necessary to comply with Discharge Specifications No. 8 and No. 9.

C. Effluent Limitations

The discharge of effluent in excess of the following limits is prohibited:

<u>Constituent</u>	<u>Units</u>	<u>30-Day Average</u>	<u>30-Day Median</u>	<u>Daily Maximum</u>
BOD	mg/l	40	--	80
Settleable Solids	ml/l	0.5		1.0
Total Coliform Organisms	MPN/100 ml	--	23	230

D. Reclamation Requirements:

1. All newly installed underground reclaimed wastewater conveyance lines shall be clearly marked as such. All aboveground reclaimed water controllers, valves, etc. shall be affixed with reclaimed water warning signs, and these and quick couplers and sprinkler heads shall be of a type, or secured in such a manner, that permits operation by authorized personnel only.
2. Public contact with reclaimed wastewater shall be precluded through such means as fences, signs, and other acceptable alternatives.
3. Areas irrigated with reclaimed water shall be managed to prevent breeding of mosquitoes. More specifically:
 - a. All applied irrigation water must infiltrate completely within 48 hours.
 - b. Ditches not serving as wildlife habitat should be maintained free of emergent, marginal, and floating vegetation.
 - c. Low-pressure and un-pressurized pipelines and ditches, which are accessible to mosquitoes, shall not be used to store reclaimed water.
4. Reclaimed water used for irrigation shall be managed to minimize both erosion and runoff from the disposal area.
5. Direct or windblown spray shall be confined to the designated reclamation area and prevented from contacting drinking water facilities.

6. The Discharger may not irrigate with effluent during periods of precipitation and for at least 24 hours after cessation of precipitation, or when soil is saturated. In addition, the Discharger may not spray irrigate when wind velocities exceed 30 mph.
7. The tailwater recapture system must be operated to capture all reclaimed water runoff, as well as any storm water runoff that occurs within 24 hours of the last application of reclaimed water.
8. To alert the public of the use of reclaimed water, signs with proper wording of sufficient size shall be placed at areas of access and around the perimeter of all areas used for effluent disposal.
9. There shall be no impoundment of reclaimed water within 1000 feet of any domestic water well or irrigation well unless it is demonstrated to the satisfaction of the Executive Officer that a shorter distance is justified.

E. Solids Disposal Requirements:

1. Collected screenings, grit, sludge, and other solids removed from liquid wastes shall be disposed of in a manner approved by the Executive Officer, and consistent with *Consolidated Regulations for Treatment, Storage, Processing, or Disposal of Solid Waste*, as set forth in Title 27, CCR, Division 2, Subdivision 1, Section 20005, et seq.
2. Storage, use and disposal of sewage sludge shall comply with existing Federal, State, and local laws and regulations, including permitting requirements and technical standards included in 40 CFR Part 503 and the Statewide General Order for the Discharge of Biosolids (Water Quality Order No. 2000-10-DWQ) (or any subsequent document which replaces Order No. 2000-10-DWQ).
3. Sludge and other solids shall be removed from ponds, clarifiers, etc. as needed to ensure optimal plant operation and adequate hydraulic capacity. Drying operations shall take place such that leachate does not impact groundwater or surface water.
4. If biosolids will be stored onsite between 15 October and 15 May of any year, then they shall be stored in a facility constructed in accordance with Class II surface impoundment or waste pile standards contained in Title 27 of the CCR. Such a facility shall be designed and maintained to prevent inundation or washout from a storm or flood with a 100 year return frequency. The Discharger shall collect any leachate or stormwater that comes in contact with the biosolids pile and return it to the wastewater treatment plant.
5. As indicated by the Discharger, beginning in December 2000, sludge from the plant shall be disposed of at an approved landfill appropriate for sludge disposal.

F. Groundwater Limitations:

The discharge in combination with other sources, shall not cause underlying groundwater to contain waste constituents in concentration statistically greater than background water quality, except for coliform bacteria. For coliform bacteria, increases shall not cause the most probable number of total coliform organisms to exceed 2.2 MPN/100 ml over any 7-day period.

G. Surface Water Limitations:

The discharge shall not cause the ephemeral drainage downstream of the storage pond to contain waste constituents in concentrations statistically greater than background (i.e., upstream) surface water quality.

H. Provisions:

1. The Discharger shall submit the following reports pursuant to Section 13267 of the California Water Code:
 - a. By **31 January 2001**, the Discharger shall submit a sludge management report. The plan shall include a description of the planned drying and storage practices.
 - b. To determine compliance with the Groundwater Limitation, the Discharger shall submit a Groundwater Monitoring Workplan by **31 March 2001** and a Groundwater Well Installation Report within **180 days** after the approval of the workplan by the Executive Officer. The Groundwater Monitoring Workplan and Monitoring Well Installation Report shall be prepared by a Registered Geologist and shall contain the information listed in Attachment B *"Items to be Included in a Monitoring Well Installation Workplan and a Monitoring Well Installation Report of Results."*
 - c. By **1 June 2001**, the Discharger shall submit a report which analyzes the problems and inadequacies in the wastewater collection, storage, and disposal system that limit the Discharger's ability to comply with Discharge Specifications No. 8 through 10 at the flow limits allowed by this Order. The report shall be prepared by a registered engineer; shall include a water balance, shall propose modifications or additions to the system to correct any inadequacies; and shall provide a proposed timeline for improvements. The timeline shall include all actions necessary to comply with CEQA and to apply for updated WDRs, if necessary.

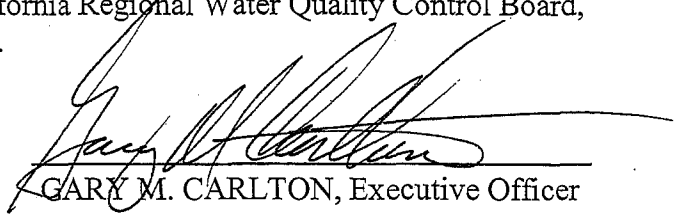
The water balance shall calculate the monthly net flow volume into the system. Net flow shall include influent, inflow/infiltration, design seasonal precipitation based on total annual precipitation using a 100-year return period (distributed monthly in accordance with historical rainfall patterns), normal evaporation rates, and measured percolation rates. The monthly net flows shall be compared with the existing storage/disposal capacity of the effluent storage pond and disposal area.

If, based on this report, system improvements are found to be necessary in order to comply with Discharge Specifications No. 8 through 10, and those improvements are not completed by **1 December 2001**, then by that date, the Discharger shall provide a contingency plan describing interim measures that will be taken in order to assure compliance with Discharge Specifications No. 8 through 10 until long-term improvements can be made.

- d. If the Monitoring and Reporting Program indicates that there are petroleum hydrocarbons in the effluent or storage pond, then within **60 days** of notification by staff, the Discharger shall submit a report showing that it has incorporated additional petroleum hydrocarbon pretreatment into its treatment system to eliminate the presence of detectable levels of petroleum hydrocarbons.
2. The Discharger shall comply with the Monitoring and Reporting Program No. 5-00-265, which is a part of this Order, and any revisions thereto as ordered by the Executive Officer.
3. The Discharger shall comply with the "Standard Provisions and Reporting Requirements for Waste Discharge Requirements", dated 1 March 1991, which are attached hereto and by reference a part of this Order. This attachment and its individual paragraphs are commonly referenced as "Standard Provision(s)."
4. At least 90 days prior to termination or expiration of any lease, contract, or agreement involving the disposal or reclamation areas, used to justify the capacity authorized herein and assure compliance with this Order, the Discharger shall notify the Board in writing of the situation and of what measures have been taken or are being taken to assure full compliance with this Order.
5. The Discharger shall submit to the Board on or before each compliance report due date the specified document, or if appropriate, a written report detailing compliance or noncompliance with the specific schedule date and task. If noncompliance is reported, then the Discharger shall state the reasons for noncompliance and shall provide a schedule to come into compliance.
6. The Discharger shall use the best practicable cost-effective control technique(s) currently available to comply with discharge limits specified in this order.
7. The Discharger shall report promptly to the Board any material change or proposed change in the character, location, or volume of the discharge.
8. In the event of any change in control or ownership of land or waste discharge facilities presently owned or controlled by the Discharger, then the Discharger shall notify the succeeding owner or operator of the existence of this Order by letter, a copy of which shall be forwarded to this office.

9. The Discharger shall comply with all conditions of this Order, including timely submittal of technical and monitoring reports as directed by the Executive Officer. Violations may result in enforcement action, including Regional Board or court orders requiring corrective action or imposing civil monetary liability, or in revision or rescission of this Order.
10. A copy of this Order shall be kept at the discharge facility for reference by operating personnel. Key operating personnel shall be familiar with its contents.
11. The Board will review this Order periodically and may revise requirements when necessary.

I, GARY M. CARLTON, Executive Officer, do hereby certify the foregoing is a full, true, and correct copy of an Order adopted by the California Regional Water Quality Control Board, Central Valley Region, on 8 December 2000.


GARY M. CARLTON, Executive Officer

Attachments
JRM:8 December 2000

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

MONITORING AND REPORTING PROGRAM NO. 5-00-265

FOR

CITY AND COUNTY OF SAN FRANCISCO
MOCCASIN SEWAGE TREATMENT PLANT
TUOLUMNE COUNTY

This Monitoring and Reporting Program (MRP) describes requirements for monitoring wastewater, effluent, the storage pond, groundwater, surface water, and biosolids. This MRP is issued pursuant to Water Code Section 13267. The Discharger shall not implement any changes to this MRP unless and until a revised MRP is issued by the Executive Officer. Specific sample station locations shall be approved by Regional Board staff prior to implementation of sampling activities.

All samples should be representative of the volume and nature of the discharge or matrix of material sampled. The time, date, and location of each grab sample shall be recorded on the sample chain of custody form.

INFLUENT MONITORING

Samples shall be collected at the same frequency and at approximately the same time as effluent samples and should be representative of the influent for the sampling period. Influent monitoring shall include, at a minimum, the following:

<u>Constituents</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u>	<u>Reporting Frequency</u>
Flow	gpd	Continuous	Daily	Monthly
BOD ₅ ¹	mg/l	Grab	Monthly	Monthly
Total Suspended Solids	mg/l	Grab	Monthly	Monthly

¹ BOD₅ denotes five-day, 20° Celsius Biochemical Oxygen Demand.

EFFLUENT MONITORING

Wastewater effluent samples shall be collected between the treatment plant outlet and the effluent storage pond. Grab samples are considered adequately composited to represent the effluent. Effluent monitoring shall include, at a minimum, the following:

<u>Constituents</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u>	<u>Reporting Frequency</u>
Flow	gpd	Continuous	Daily	Monthly
BOD ₅ ¹	mg/l	Grab	Bi-weekly	Monthly
Total Settleable Solids	ml/l•hr	Grab	Bi-weekly	Monthly
Total Coliform Organisms	MPN/100 ml	Grab	Bi-weekly	Monthly
Total Suspended Solids	mg/l	Grab	Monthly	Monthly
Total Dissolved Solids	mg/l	Grab	Monthly	Monthly
Nitrates as Nitrogen	mg/l	Grab	Monthly	Monthly
TPH-G, D, MO	ug/l	Grab	Quarterly	Quarterly
BTEX	ug/l	Grab	Quarterly	Quarterly

¹ BOD₅ denotes five-day, 20° Celsius Biochemical Oxygen Demand.

STORAGE POND MONITORING

Samples shall be collected from an established sampling station located in an area that will provide a sample representative of the wastewater in the pond. Freeboard will be measured vertically from the surface of the pond water to the lowest elevation of the berm. Freeboard shall be measured to the nearest 0.25 feet. Pond monitoring shall include, at a minimum, the following:

<u>Constituent</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u>	<u>Reporting Frequency</u>
Freeboard	Feet	Measurement	Daily	Monthly
PH	PH units	Grab	Weekly	Monthly
Dissolved Oxygen	mg/l	Grab	Weekly	Monthly
TPH-G, D, MO	ug/l	Grab	Quarterly	Quarterly
BTEX	ug/l	Grab	Quarterly	Quarterly

SPRAYFIELD MONITORING

The sprayfield shall be monitored on a daily basis when spray irrigation is being conducted. Sprayfield monitoring shall include, at a minimum, the following:

<u>Parameter</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u>	<u>Reporting Frequency</u>
Flow	gpd	Continuous	Daily	Monthly
Runoff	gpm	Estimation	Daily	Monthly

GROUNDWATER MONITORING

Prior to construction of any groundwater monitoring wells, the Discharger shall submit plans and specifications to the Board for review and approval. Once installed, all new wells shall be added to the MRP and shall be sampled and analyzed according to the schedule below.

Prior to sampling, the groundwater elevations shall be measured and the wells shall be purged at least three well volumes until pH and electrical conductivity have stabilized. Depth to groundwater shall be measured to the nearest 0.01 feet. Samples shall be collected using standard EPA methods. Groundwater monitoring shall include, at a minimum, the following:

<u>Constituent</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling and Reporting Frequency</u>
Groundwater elevation	Feet	Measurement	Quarterly
Total Dissolved Solids	mg/l	Grab	Quarterly
Ammonia as Nitrogen	mg/l	Grab	Quarterly
Nitrates as Nitrogen	mg/l	Grab	Quarterly
Total Coliform Organisms	MPN/100 ml	Grab	Quarterly

SURFACE WATER MONITORING

The Discharger shall establish two sampling stations within the seasonal drainage which contains the effluent storage pond: one station (S-1) shall be located 30 feet upstream of the pond and one station (S-2) approximately 30 feet downstream of the pond. Surface water samples shall be collected during those months when water is present in the drainage. Surface water samples shall be analyzed for the following:

<u>Constituent</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling and Reporting Frequency</u>
Total Dissolved Solids	mg/l	Grab	Monthly ¹
Ammonia as Nitrogen	mg/l	Grab	Monthly ¹
Nitrates as Nitrogen	mg/l	Grab	Monthly ¹
Total Coliform Organisms	MPN/100ml	Grab	Monthly ¹
TPH-G, D, MO	ug/l	Grab	Monthly ¹
BTEX	ug/l	Grab	Monthly ¹

¹ When surface water is present.

² Using a minimum of 25 tubes or five dilutions.

BIOSOLIDS MONITORING

The Discharger shall keep records regarding the quantity of biosolids generated by the treatment processes; any sampling and analytical data; the quantity of biosolids stored on site; and the quantity removed for disposal. The records shall also indicate steps taken to reduce odor and other nuisance conditions. Records shall be stored onsite and available for review during inspections.

If biosolids are transported off-site for disposal, then the Discharger shall submit records identifying the hauling company, the amount of biosolids transported, and the location of disposal. If biosolids are disposed of onsite, then the Discharger shall submit the annual report information as contained in the Statewide General Order for the Discharge of Biosolids (Water Quality Order No. 2000-10-DWQ) (or any subsequent document which replaces Order No. 2000-10-DWQ).

All records shall be submitted as part of the Annual Monitoring Report.

REPORTING

In reporting monitoring data, the Discharger shall arrange the data in tabular form so that the date, sample type (e.g., effluent, pond, etc.), and reported analytical result for each sample are readily discernible. The data shall be summarized in such a manner to clearly illustrate compliance with waste discharge requirements and spatial or temporal trends, as applicable. The results of any monitoring done more frequently than required at the locations specified in the Monitoring and Reporting Program shall be reported in the next scheduled monitoring report.

As required by the California Business and Professions Code Sections 6735, 7835, and 7835.1, all Groundwater Monitoring Reports shall be prepared under the direct supervision of a Registered Engineer or Geologist and signed by the registered professional.

A. Monthly Monitoring Reports

Monthly reports shall be submitted to the Regional Board on the **1st day of the second month following sampling** (i.e. the January Report is due by 1 March). At a minimum, the reports shall include:

1. Results of influent, effluent, storage pond, and surface water monitoring. Data shall be presented in tabular format.
2. A comparison of monitoring data to the discharge specifications and an explanation of any violation of those requirements.
3. If requested by staff, copies of laboratory analytical report(s).
4. A calibration log verifying weekly calibration of field monitoring instruments (DO, pH, EC, etc. meters) used to collect reported data.

5. A discussion of compliance and the corrective action taken, as well as any planned or proposed actions needed to bring the discharge into full compliance with the waste discharge requirements.

B. Quarterly Monitoring Reports

The Discharger shall establish a quarterly sampling schedule for groundwater monitoring such that samples are obtained approximately every three months. Quarterly monitoring reports shall be submitted to the Board by the **1st day of the second month after the quarter** (i.e. the January-March quarter is due by May 1st) each year. The Quarterly Report shall include the following:

1. Results of groundwater monitoring. The results of regular monthly monitoring reports for March, June, September and December may be incorporated into their corresponding quarterly monitoring report.
2. Field logs shall be submitted for each well, documenting depth to groundwater; parameters measured before, during, and after purging; method of purging; calculation of casing volume; and total volume of water purged.
3. Calculation of groundwater elevations, an assessment of groundwater flow direction and gradient on the date of measurement, comparison of previous flow direction and gradient data, and discussion of seasonal trends if any.
4. A narrative discussion of the analytical results for all media and locations monitored including spatial and temporal trends, with reference to summary data tables, graphs, and appended analytical reports (as applicable).
5. A comparison of monitoring data to the discharge specifications, groundwater limitations and surface water limitations, and explanation of any violation of those requirements.
6. Summary data tables of historical and current water table elevations and analytical results.
7. A scaled map showing relevant structures and features of the facility, the locations of monitoring wells and any other sampling stations
8. Copies of laboratory analytical report(s) for groundwater, if requested by staff.

C. Annual Report

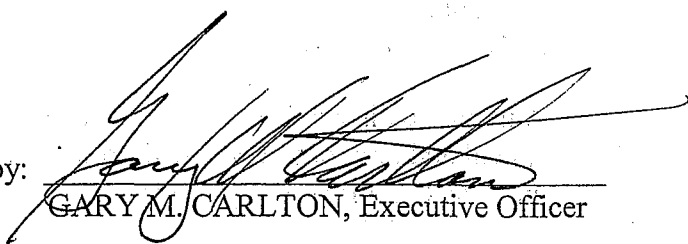
An Annual Report shall be prepared as the fourth quarter monitoring report. The Annual Report will include all monitoring data required in the monthly/quarterly schedule. The Annual Report shall be submitted to the Regional Board by **1 February** of each year:

1. If requested by staff, tabular and graphical summaries of all data collected during the year.
2. Data for monitoring and analysis performed on an annual basis (i.e., biosolids monitoring).

3. An evaluation of the performance of the wastewater treatment system and effluent storage pond, as well as a forecast of the flows anticipated in the next year.
4. An evaluation of the groundwater quality beneath the effluent storage pond.
5. Summary of information on the disposal of all biosolids including volume removed and location of disposal, and analytical results.
6. Whether the Discharger anticipates removing biosolids in the coming year, and if so, an anticipated schedule for cleaning, drying, and disposal;
7. A discussion of compliance and the corrective action taken, as well as any planned or proposed actions needed to bring the discharge into full compliance with the waste discharge requirements.
8. A discussion of any data gaps and potential deficiencies/redundancies in the monitoring system or reporting program.

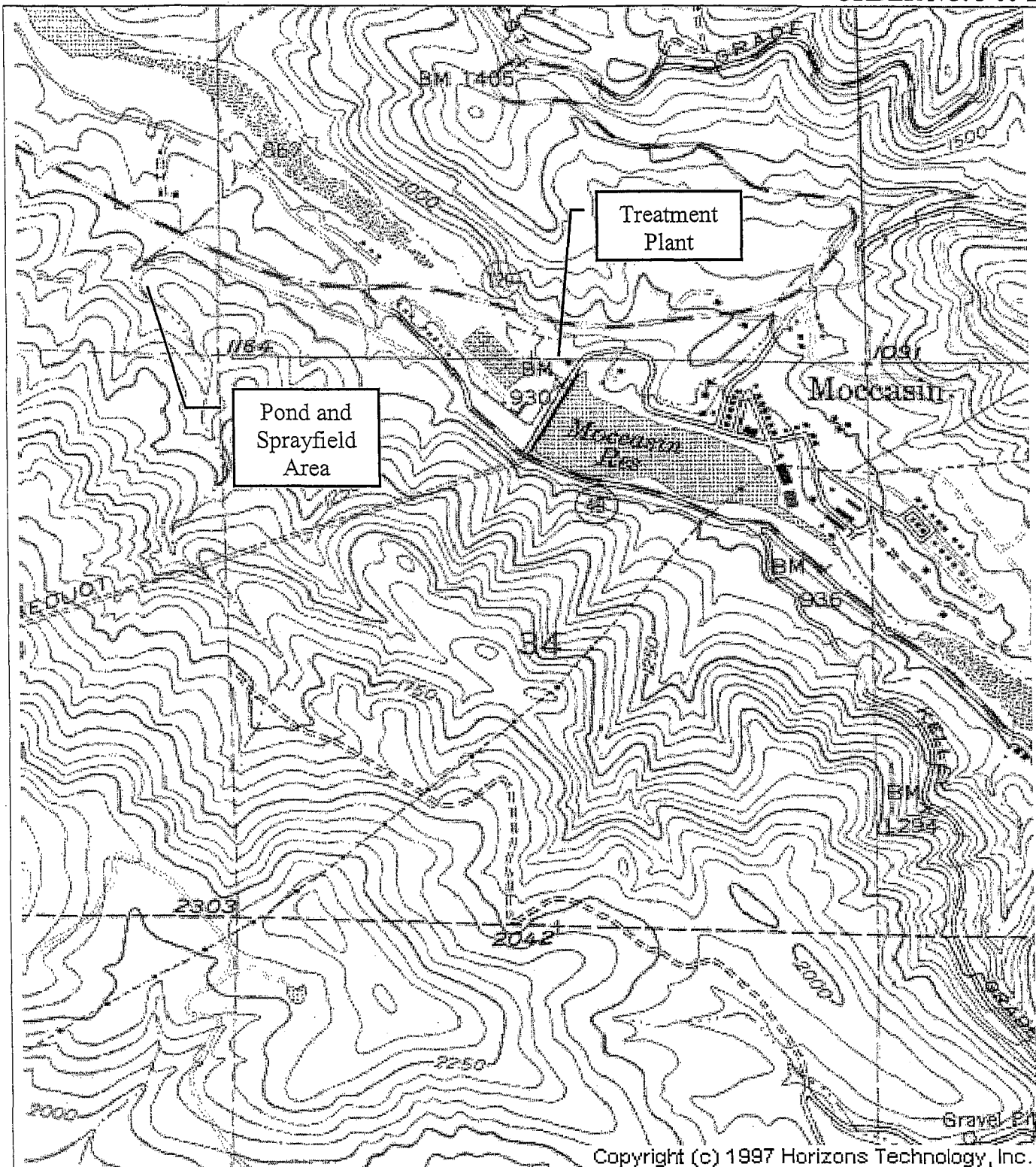
The Discharger shall implement the monitoring program for treatment plant operations, influent, effluent, storage pond, sprayfield, and biosolids monitoring as of the date of this Order. The Discharger shall implement groundwater monitoring when approved groundwater monitoring facilities are in place and operational.

Ordered by:


GARY M. CARLTON, Executive Officer

8 December 2000

(Date)

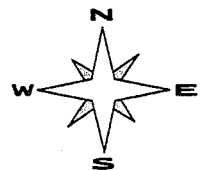


Drawing Reference:

U.S.G.S TOPOGRAPHIC MAP
7.5 MINUTE QUADRANGLE
Photorevised 1964

SITE LOCATION MAP

Moccasin Sewage Treatment Plant
Tuolumne County



approx. scale
1 in. = 1,200 ft.



California Regional Water Quality Control Board

Central Valley Region

Steven T. Butler, Chair



Gray Davis
Governor

Winston H. Hickox
Secretary for
Environmental
Protection

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ATTACHMENT B

ITEMS TO BE INCLUDED IN A MONITORING WELL INSTALLATION WORKPLAN AND A MONITORING WELL INSTALLATION REPORT OF RESULTS

Prior to installation of groundwater monitoring wells, the Discharger shall submit a workplan containing the minimum listed information. Wells may be installed after staff approve the workplan. Upon installation of the monitoring wells, the Discharger shall submit a report of results, as described below. All workplans and reports must be signed by a registered geologist, certified engineering geologist, or civil engineer registered or certified by the State of California.

Monitoring Well Installation Workplan

A. General Information:

- Monitoring well locations and rationale
- Survey details
- Equipment decontamination procedures
- Health and safety plan
- Topographic map showing any existing monitoring wells, proposed wells, waste handling facilities, utilities, and other major physical and man-made features.

B. Drilling Details: describe drilling and logging methods

C. Monitoring Well Design:

- Casing diameter
- Borehole diameter
- Depth of surface seal
- Well construction materials
- Diagram of well construction
- Type of well cap
- Size of perforations and rationale
- Grain size of sand pack and rationale
- Thickness and position of bentonite seal and sand pack
- Depth of well, length and position of perforated interval

D. Well Development:

- Method of development to be used
- Method of determining when development is complete
- Method of development water disposal

- E. Surveying Details: discuss how each well will be surveyed to a common reference point
- F. Soil Sampling (if applicable):
 - Cuttings disposal method
 - Analyses to be run and methods
 - Sample collection and preservation method
 - Intervals at which soil samples are to be collected
 - Number of soil samples to be analyzed and rationale
 - Location of soil samples and rationale
 - QA/QC procedures
- G. Well Sampling:
 - Minimum time after development before sampling (48 hours)
 - Well purging method and amount of purge water
 - Sample collection and preservation method
 - QA/QC procedures
- H. Water Level Measurement:
 - The elevation reference point at each monitoring well shall be within 0.01 foot. Ground surface elevation at each monitoring well shall be within 0.1 foot. Method and time of water level measurement shall be specified.
- I. Proposed time schedule for work.

Monitoring Well Installation Report of Results

- A. Well Construction:
 - Number and depth of wells drilled
 - Date(s) wells drilled
 - Description of drilling and construction
 - Approximate locations relative to facility site(s)
 - A well construction diagram for each well must be included in the report, and should contain the following details:
 - Total depth drilled
 - Depth of open hole (same as total depth drilled if no caving occurs)
 - Footage of hole collapsed
 - Length of slotted casing installed
 - Depth of bottom of casing
 - Depth to top of sand pack
 - Thickness of sand pack
 - Depth to top of bentonite seal
 - Thickness of bentonite seal
 - Thickness of concrete grout
 - Boring diameter

- Casing diameter
- Casing material
- Size of perforations
- Number of bags of sand
- Well elevation at top of casing
- Depth to ground water
- Date of water level measurement
- Monitoring well number
- Date drilled
- Location

B. Well Development:

- Date(s) of development of each well
- Method of development
- Volume of water purged from well
- How well development completion was determined
- Method of effluent disposal
- Field notes from well development should be included in report.

C. Well Surveying: provide reference elevations for each well and surveyor's notes

D. Water Sampling:

- Date(s) of sampling
- How well was purged
- How many well volumes purged
- Levels of temperature, EC, and pH at stabilization
- Sample collection, handling, and preservation methods
- Sample identification
- Analytical methods used
- Laboratory analytical data sheets
- Water level elevation(s)
- Groundwater contour map

E. Soil Sampling (if applicable):

- Date(s) of sampling
- Sample collection, handling, and preservation method
- Sample identification
- Analytical methods used
- Laboratory analytical data sheets

WSW

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26 October 1999

INFORMATION SHEET

WASTE DISCHARGE REQUIREMENTS ORDER NO. 5-00-265
CITY AND COUNTY OF SAN FRANCISCO
MOCCASIN SEWAGE TREATMENT PLANT
TUOLUMNE COUNTY

The City and County of San Francisco (hereafter Discharger) operates a wastewater treatment plant at the Moccasin Powerhouse and the Hetch Hetchy Water and Power Administrative complex. Wastewater originates from the complex that contains 44 residences, a maintenance shop, the powerhouse (domestic waste only), and the administrative offices. The wastewater treatment plant serves approximately 50 to 60 permanent residents, 70 to 80 workers per day using the facility during the week, and 25 to 30 workers from remote locations that periodically use the facilities. The current dry weather flows are approximately 20,000 gallons per day (gpd), with peak wet weather flows reaching approximately 70,000 gpd.

The treatment system consists of an extended aeration package plant, with disposal by means of spray irrigation. Effluent from the treatment plant is pumped from the chlorine contact tank through a force main to the effluent storage pond. The effluent storage pond is in a seasonal drainage, with an earthen dam to increase capacity. Upslope storm water is diverted beyond the pond by means of a 48-inch pipe that passes under the pond and dam. The capacity of the pond while maintaining two feet of freeboard is approximately 480,000 gallons.

The Discharger has no record of overflows from the pond; however, a water balance for the effluent storage and disposal system indicates that there is not enough storage capacity in the pond to assure that two feet of freeboard is maintained during the wet season. Effluent is sprayed onto the 10-acre sprayfield. A diversion ditch directs runoff from the spray field. Runoff can be directed back to the pond, or can be routed around the pond and into the seasonal drainage below the pond.

The Monitoring and Reporting Program includes requirements for monitoring of influent and effluent, the storage pond, the sprayfield, surface water, groundwater, and biosolids. In order to determine compliance with the groundwater limitation contained herein, these WDRs contain a time schedule for development and implementation of a groundwater monitoring program.

JRM 12/8/00

